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Dr Aleksandar Pasquini, Professor Rajesh Vasa
Deakin Applied Artificial Intelligence Initiative
Deakin University
221 Burwood Highway
Burwood, VIC 3125

Dear Dr Pasquini and Professor Vasa,

RE: Application for the Smarter AI Agent Routing Through Task Understanding research internship

I am writing to apply for the **Smarter AI Agent Routing Through Task Understanding** opportunity, advertised through Deakin's **High Achievers' Internship Program with the Applied Artificial Intelligence Initiative**. What drew me to this project is a question I had never thought to ask about my own use of AI: how does a system know a task needs a stronger model before it has even tried?

I am completing a Bachelor of Software Engineering (Honours), expected to graduate in 2028, with an 81.33 WAM and scores of 100 in Backend Development and Data Structures and Algorithms. For my Embedded Systems unit I built a smart clothesline robot whose rain sensor was its least reliable component, and a wrong reading in either direction could make the robot take the wrong action. Rather than let one uncertain reading decide, the Python and Flask backend I built for it on a Raspberry Pi checks the Open-Meteo forecast and sends the robot to shelter when the sensor is unsure but rain is likely. That is a small version of your brief's point that a system must recognise when its information is not enough to decide.

Your brief also describes requests that are "more complex than they initially appear", and two of my hardest problems were exactly that. In my Robot Controller API, which scored 100/100 and was showcased at Deakin Open Day, adding rollback sounded like undoing the last job, until I found that undoing jobs out of order leaves the robot somewhere it never was. On the physical robot, random resets looked like a code bug and turned out to be a power problem I could only prove from its telemetry. I could not tell either was hard until I was inside it. Designing the bugs seeded into a Deakin Software Engineering Club workshop app for around 30 students taught me the reverse: difficulty has to be set deliberately, which is what your dataset of clear to ambiguous tasks needs.

What excites me about this project is that I am the user your brief describes. I use Claude Code every day and always choose the strongest model, because as one student I have never run out of credits, and I could not say which tasks actually needed it. At the scale of a multi-agent system, that choice is no longer free: every routing decision becomes a trade-off between quality, speed and cost. I have also learnt that context decides quality: I keep key facts in a CLAUDE.md file and organise my folders so Claude looks in the right place for each task. I am manually supplying the kind of context your router needs to extract from a request, and I am curious which signals are useful enough to change its decision, and how conformal prediction behaves when uncertainty comes from an unfamiliar task rather than an ambiguous request. I have not used it yet, and it is where I expect to learn the most.

I would bring a habit of designing for uncertainty, hands-on Python and backend experience, and real curiosity about when a system should trust its own choice. I am available on campus at Burwood two days per week for the full ten weeks. Please find my resume and academic transcript attached. I would welcome the opportunity to discuss my application in an interview. Thank you for your consideration.

Kind regards,

Manit Khera